

Kuber Sarwal

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science, Artificial Intelligence and Systems & Architecture

May 2028

- **GPA: 3.86**
- **Coursework:** Data Structures and Algorithms, Computer Organization and Programming, Discrete Math, Linear Algebra, Differential Equations, Object-Oriented Programming, Modern Physics

EXPERIENCE

Machine Learning Engineer Intern – GNC

Irvine, CA

Terran Orbital Corporation, a Lockheed Martin Company

Summer 2026

- Custom-built and trained a **U-Net CNN** on proprietary night-sky imagery to detect star centroids under high-speed maneuvers via **star centroid maps**, achieving **50 arcsecond pointing accuracy** at a **5 deg/s slew rate** – a **3x improvement** over baseline.
- Implemented **QUEST algorithm** consuming CNN-extracted centroids for real-time **quaternion-based attitude estimation**; optimized **low-latency** inference on **NVIDIA Jetson** edge hardware with **OpenCV**, profiling **GPU/ML accelerator microarchitecture** to cut latency **40%**.
- Developed a **real-time synthetic star-streak simulation environment** to test/validate the CNN/QUEST pipeline pre-deployment; validated attitude estimates against IMU telemetry across **12+ hardware-in-the-loop tests**, informing model tuning decisions ahead of **production** deployment.

Machine Learning Engineer, VIP – AI MakerSpace Nexus VI

Atlanta, GA

Georgia Institute of Technology (Sponsored by NVIDIA)

Jan. 2026 – Present

- Built **StudyBuddy**, an end-to-end **AI-powered application (OCR, NLP, RAG)** with React/Llama 3/Ollama; orchestrated **LLM workflows** via **LangChain/GPT** for adaptive learning; used a **SQL (SQLite)** data pipeline for progress tracking and content tagging.
- Owned the pipeline from **data ingestion to deployment**: fine-tuned **ChatGT** on **Llama 3**, trained multimodal **deep learning** models on **NVIDIA PACE ICE** with **CUDA GPU** acceleration and **MLflow** tracking, achieving **70%** faster comprehension across **50+ datasets**.

Research Intern

Camden, NJ

Rutgers University

Jan. 2024 – Jan. 2025

- Analyzed algorithmic bias in redlining via **geospatial analysis** of **SQL**-queried HMDA data (25M+ samples); built **predictive models** (Random Forest, SVM, Neural Nets) in **PyTorch/TensorFlow/Spark**; reduced bias **59%** via **AI Fairness 360**, published at **IEEE BigData 2024**.

Machine Learning Engineer Intern

Montgomery, NJ

Learnroll LLC

Dec. 2024 – Mar. 2025

- Labeled **10K+ frames** using **CVAT** and trained a **YOLOv8** model to analyze CPR keypoints; applied **OpenCV** preprocessing; containerized via **Docker** and deployed to a **low-latency production** environment on **AWS EC2**.

PROGRAMMING PROJECTS

Pebble | React, Flask, Python, OpenCV, YOLO, LangChain, Google Gemini API

Oct. 2025

- Built an AI study companion tracking gaze, posture, and phone usage via **OpenCV** and **YOLO (94% accuracy)**; deployed an **LLM agent** with **LangChain/Vapi** for real-time alerts via a **REST API** connecting **React/Flask**.

Perfect Punch - BDBI@GT | Python, Flask, React, KNN, Neural Networks, Random Forests, PyGSep. 2025 – Present

- Built an AI boxing assistant classifying punches via **active learning (QBC)** with KNNs, Neural Networks, and Random Forests (**96% accuracy**); modeled skeletal keypoints as graphs using **PyTorch Geometric** for **graph ML**; integrated **LLM coaching agent** via **Flask REST API/React**.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML/CSS, SQL, NoSQL, C++, C#, R, Go, Rust, TypeScript, MATLAB

Frameworks: React, Node, Flask, Angular, Django, FastAPI, AWS, Apache Spark, Kafka, LangChain, Spring

Developer Tools: Git, GCP, SageMaker, Azure, MongoDB, CUDA, MLflow, Airflow, Docker, Kubernetes, MCP Servers

Libraries: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-learn, OpenCV, PyG, DGL, ONNX, TensorRT,

GeoPandas, GDAL, Rasterio, CoreFlow